

CITS3002: Lab 4

Exercise 1:

Consider a shared communication channel using ALOHA protocols where the frame time is 10 ms, there are 10 terminals, and each terminal generates frames (including retransmissions) with a mean rate of 1 frame per second.

- a) What is the probability that exactly one frame is generated during one frame time (10 ms) across all terminals?
- b) Write a Python program to simulate and compute the throughput of both Pure ALOHA and Slotted ALOHA under the above conditions.
- c) Write a Python program to evaluate the performance of Pure ALOHA and Slotted ALOHA by varying G from 0.1 to 1.5 in increments of 0.1. Then, draw a chart that compares the throughput of Pure ALOHA and Slotted ALOHA as a function of G , with both curves displayed on the same graph.
- d) What are the maximum throughput values achievable for Pure ALOHA and Slotted ALOHA?

Exercise 2:

Consider a classic shared-medium Ethernet network using the CSMA/CD protocol. Assume that the channel is initially idle, and two terminals have a frame ready to transmit at the same time.

- a) Describe what happens when both terminals attempt to transmit. Explain the sequence of events step by step.
- b) After a collision, the terminals apply the binary exponential backoff algorithm. Write a Python program that simulates this process and determines after how many retransmissions each frame is successfully delivered.

Exercise 3:

Consider a classic shared-medium Ethernet network. Suppose that there are 20 stations, and each station transmits in a contention slot with a probability of 0.4.

- a) Calculate the probability that one station transmits successfully in a contention slot.
- b) Calculate the mean number of slots per contention interval.
- c) Assume that the frame transmission time is $200\ \mu\text{s}$ and the one-way propagation delay is $\tau = 25\ \mu\text{s}$. Given that each contention slot has duration 2τ , calculate the channel efficiency.